RHCSA SA1 Event
Practical Report

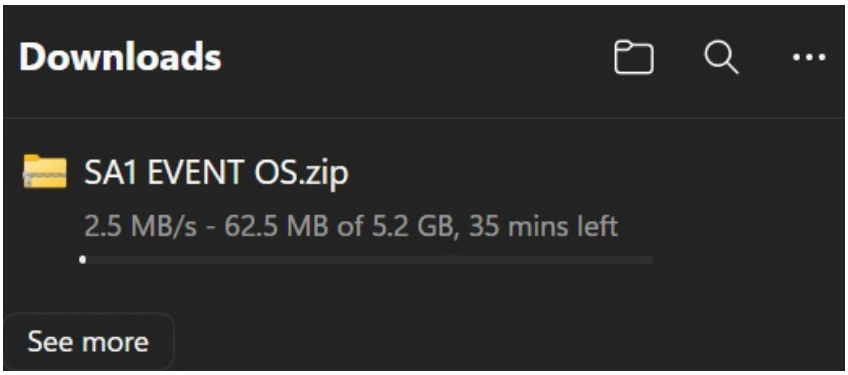
10-09-26

Submitted by
Abdul Janeeb K
11:30 Batch
Cloud & DevOps

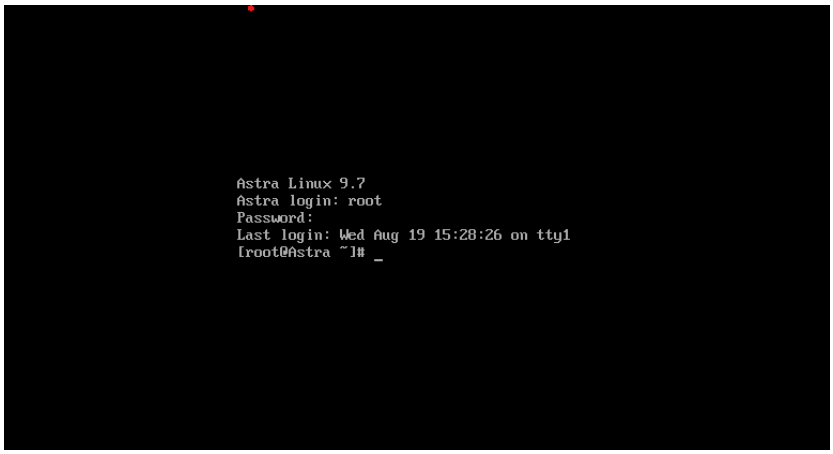
Phase 1 – Operating System Setup & SSH Connectivity

Step 1: Download and Login

I downloaded the SA1 event OS zip file from Google Drive. It was around 5.2 GB, so it took a while to finish.



Once it was done, I extracted the file and added it to VMware, then logged in using the root credentials given in the doc.

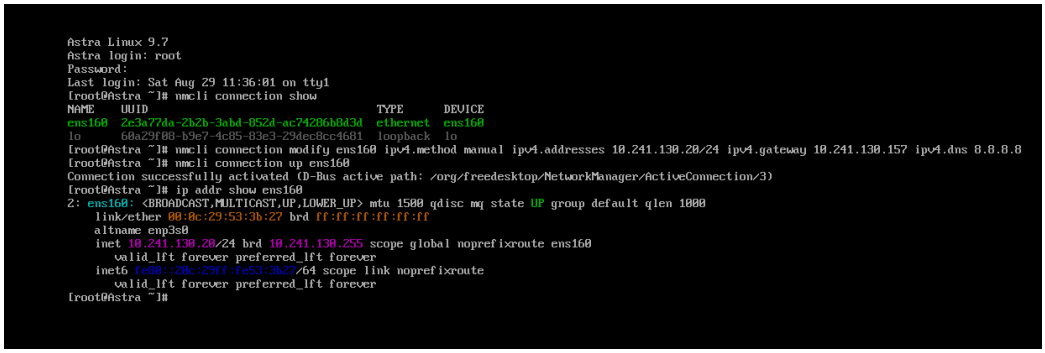


Step 2: Configure IP Address Manually

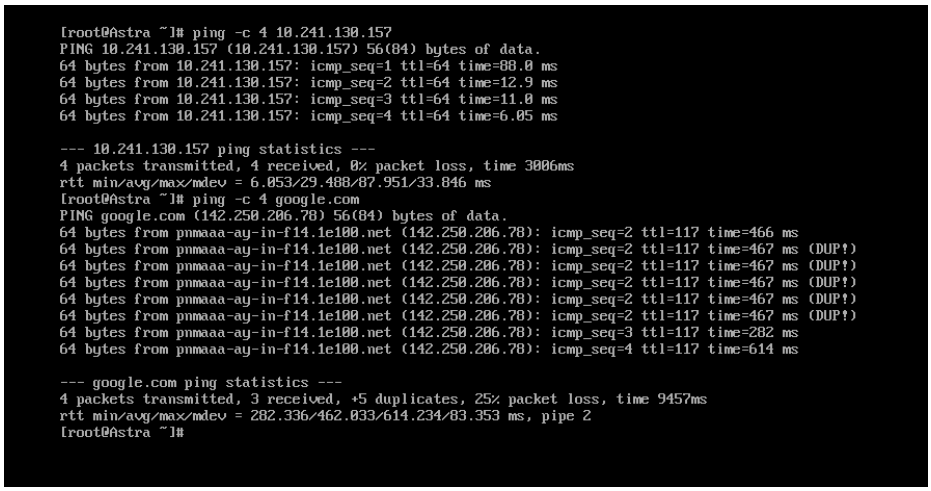
Before assigning an IP to the VM, I checked my host machine's network details using ipconfig on Windows, so I'd know the subnet and gateway and could pick a free IP in the same range.



I used nmcli to set a static IP manually on ens160, kept the same gateway as the host, and set DNS to 8.8.8.8. Then I checked it with ip addr show.

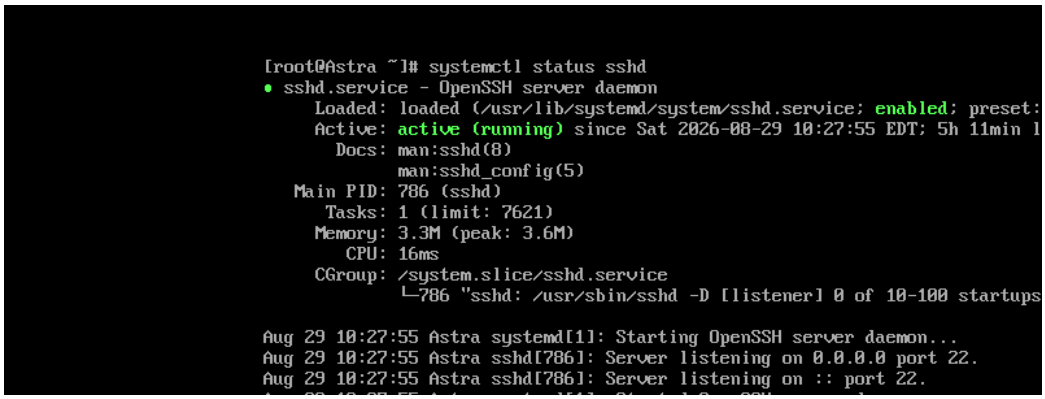


To make sure the network was actually working, I pinged the gateway and google.com.



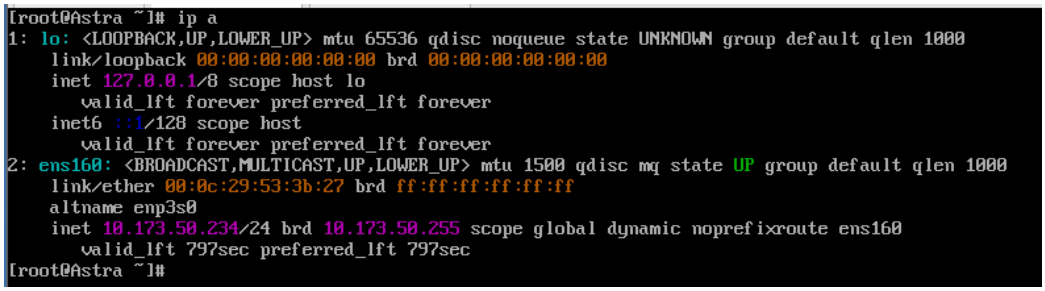
Step 3: Configure SSH

I checked the SSH service with systemctl status sshd – it was already enabled and active/running, so I didn't need to install anything.



Step 4: Connect Using PuTTY

Before opening PuTTY, I checked the IP again with ip a to confirm ens160 still had 10.173.50.234/24.

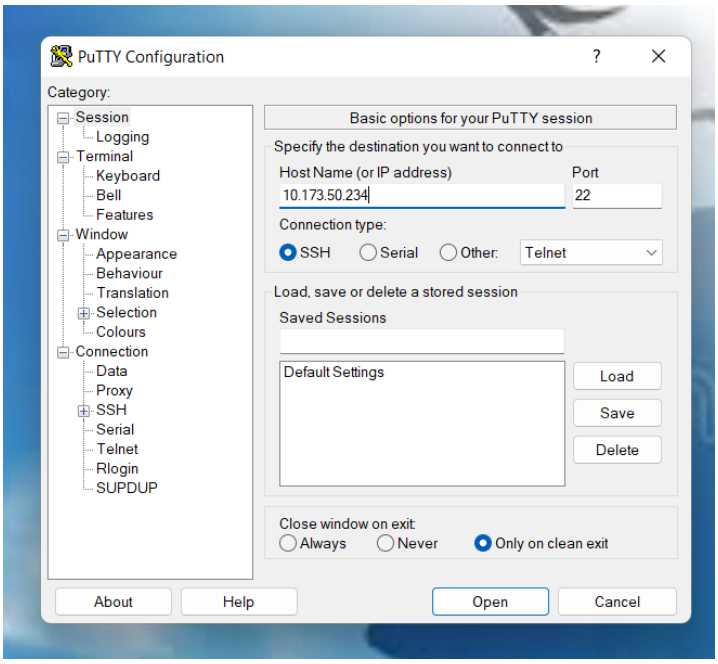


I also checked systemctl status sshd once more to make sure the service was active and listening on port 22.

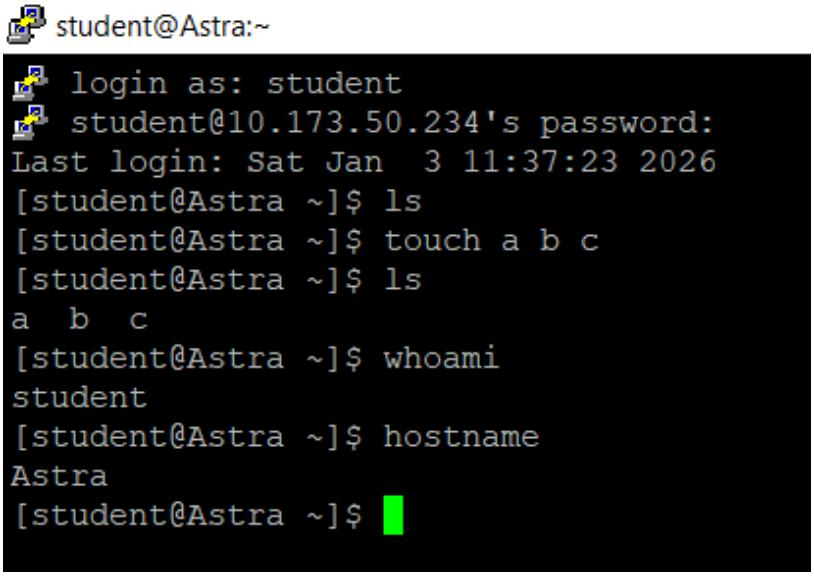
```
[root@Astra ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-09-09 06:13:06 EDT; 1h 50min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 783 (sshd)
      Tasks: 1 (limit: 7621)
     Memory: 3.3M (peak: 3.6M)
        CPU: 21ms
     CGroup: /system.slice/sshd.service
             └─783 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 09 06:13:06 Astra systemd[1]: Starting OpenSSH server daemon...
Sep 09 06:13:06 Astra sshd[783]: Server listening on 0.0.0.0 port 22.
Sep 09 06:13:06 Astra sshd[783]: Server listening on :: port 22.
Sep 09 06:13:06 Astra systemd[1]: Started OpenSSH server daemon.
[root@Astra ~]# _
```

Then I opened PuTTY on the Windows host and entered the VM's static IP (10.173.50.234) with port 22.



It logged in through SSH from the Windows host. To check the session was actually live I ran ls, touch a b c, whoami, and hostname, and they all worked.



Phase 2 – User Management, Permissions, Monitoring & Graphical Environment

Task 1: Create and Configure User Account

I created my own user account with useradd abduljaneeb, set a password with passwd, then checked it was there using id and tail /etc/passwd.

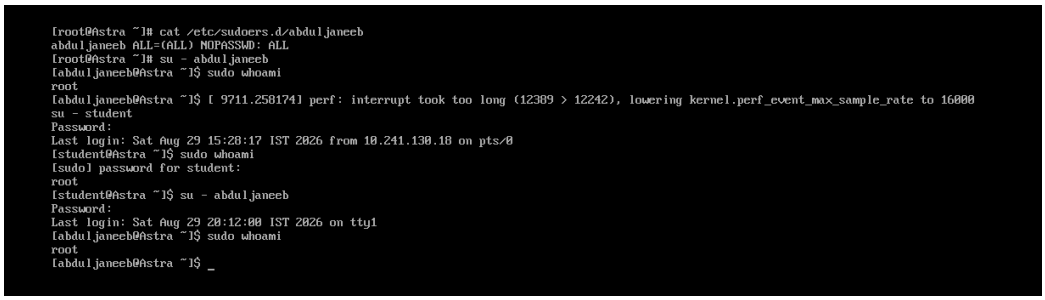


For sudo access without a password, I added a sudoers drop-in file at /etc/sudoers.d/abduljaneeb with this line instead of editing /etc/sudoers directly:

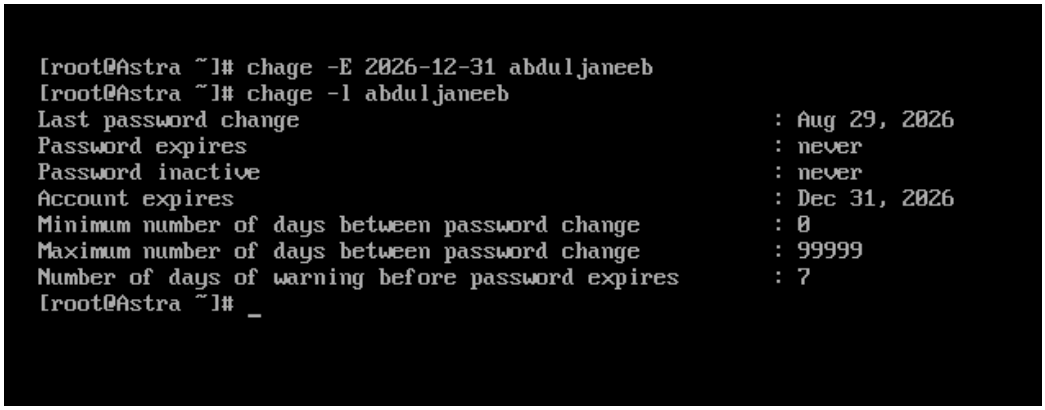
abduljaneeb ALL=(ALL) NOPASSWD: ALL



I switched to abduljaneeb and ran sudo whoami to check it. It returned root without asking for a password, so NOPASSWD was working.



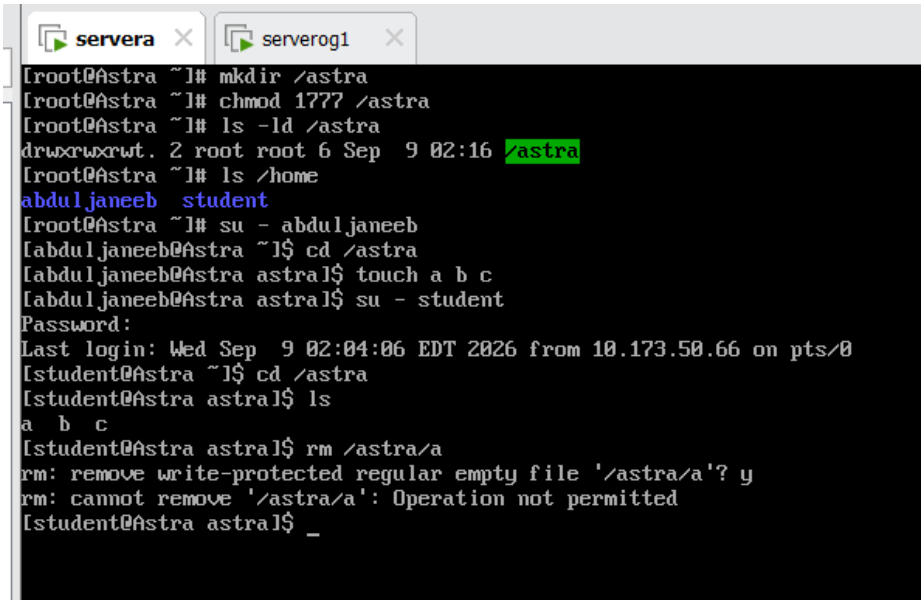
Since this account was only for the practical, I set it to expire on Dec 31, 2026 using chage -E, then checked it with chage -l abduljaneeb.



Task 2: Create and Configure the /astra Directory

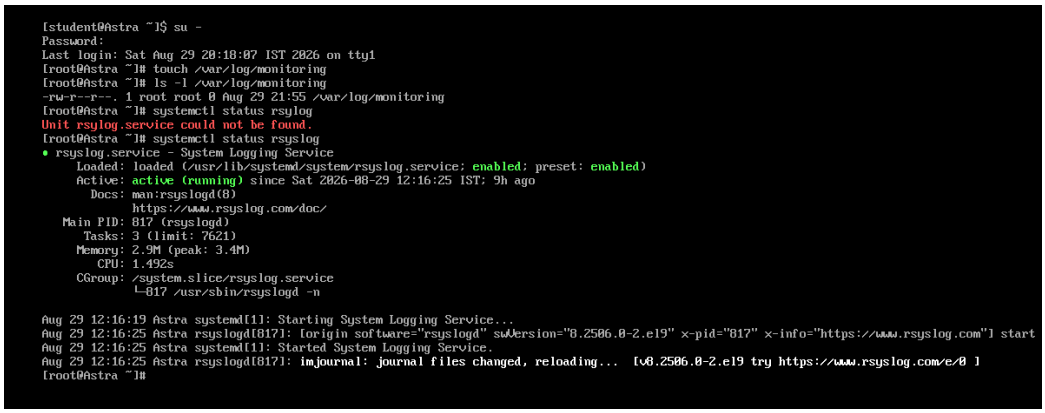
I created /astra and ran chmod 1777 on it – the sticky bit – so any user could create files inside, but only the owner of a file could delete it.

I checked the permissions with ls -ld /astra, then logged in as abduljaneeb and created a file, and as student tried to delete it. student got “Operation not permitted”, so the sticky bit was doing its job.



Task 3: Monitoring and Activity Logging

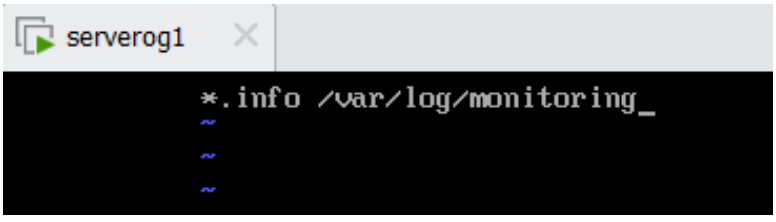
I created the log file with touch /var/log/monitoring and checked rsyslog was active and running.



Then I created a config file at /etc/rsyslog.d/monitoring.conf.



Inside it I added the rule *.info /var/log/monitoring, so info-level activity gets logged to this file.



I restarted rsyslog and tested it with the logger command. The test message showed up in /var/log/monitoring, so the logging was working.

```
[root@Astra ~]# systemctl restart rsyslog
[root@Astra ~]# logger "This is a Test Message From Janeeb"
[root@Astra ~]# tail /var/log/monitoring
Sep  9 03:00:43 Astra rsyslogd[10963]: imjournal: journal files changed, reloading..
Sep  9 03:01:01 Astra CROND[10969]: (root) CMD (run-parts /etc/cron.hourly)
Sep  9 03:01:01 Astra run-parts[10972]: (/etc/cron.hourly) starting 0anacron
Sep  9 03:01:01 Astra anacron[10982]: Anacron started on 2026-09-09
Sep  9 03:01:01 Astra anacron[10982]: Will run job `cron.daily' in 34 min.
Sep  9 03:01:01 Astra anacron[10982]: Will run job `cron.weekly' in 54 min.
Sep  9 03:01:01 Astra anacron[10982]: Jobs will be executed sequentially
Sep  9 03:01:01 Astra run-parts[10984]: (/etc/cron.hourly) finished 0anacron
Sep  9 03:01:01 Astra CROND[10968]: (root) CMDEND (run-parts /etc/cron.hourly)
Sep  9 03:02:03 Astra root[10985]: This is a Test Message From Janeeb
[root@Astra ~]# _
```

Task 4: Configuring Graphical Environment

I ran yum groupinstall 'server with GUI', but it failed the first time – it couldn't resolve the repo mirror host. I fixed the network by setting it back to auto with nmcli and reconnecting, and the IP came back fine. I ran the group install again and it started downloading packages.

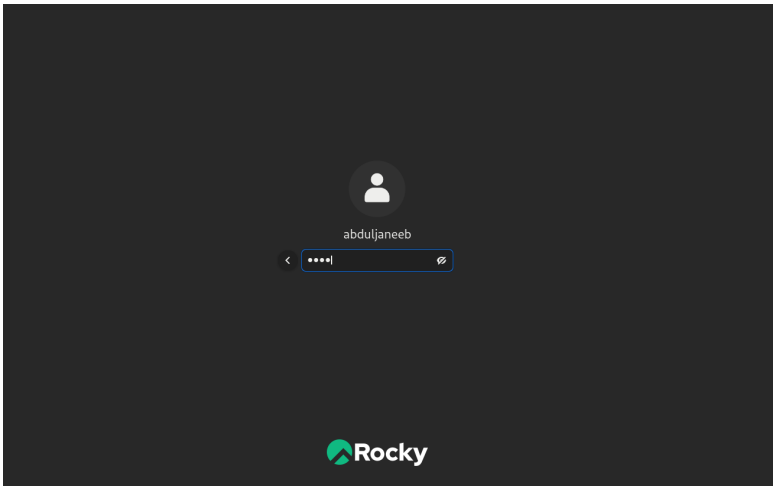
```
[root@Astra ~]# yum groupinstall -y 'server with GUI'
Astra Linux 9 - BaseOS
Errors during downloading metadata for repository 'baseos':
  - Curl error (6): Couldn't resolve host name for https://mirrors.rockylinux.org/mirrorlist?arch=x86_64&repo=BaseOS-9 [Coul
ux.org]
Error: Failed to download metadata for repo 'baseos': Cannot prepare internal mirrorlist: Curl error (6): Couldn't resolve h
nux.org/mirrorlist?arch=x86_64&repo=BaseOS-9 [Could not resolve host: mirrors.rockylinux.org]
[root@Astra ~]# nmcli connection modify ens160 ipv4.method auto ipv4.addresses "" ipv4.gateway "" ipv4.dns ""
[root@Astra ~]# nmcli connection down ens160
Connection 'ens160' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
[root@Astra ~]# nmcli connection up ens160
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
[root@Astra ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:53:3b:27 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.141.128/24 brd 192.168.141.255 scope global dynamic noprefixroute ens160
       valid_lft 1792sec preferred_lft 1792sec
    inet6 fe80::20c:29ff:fe53:3b27/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@Astra ~]# yum groupinstall -y 'server with GUI'
```

The GUI packages installed without any more issues.

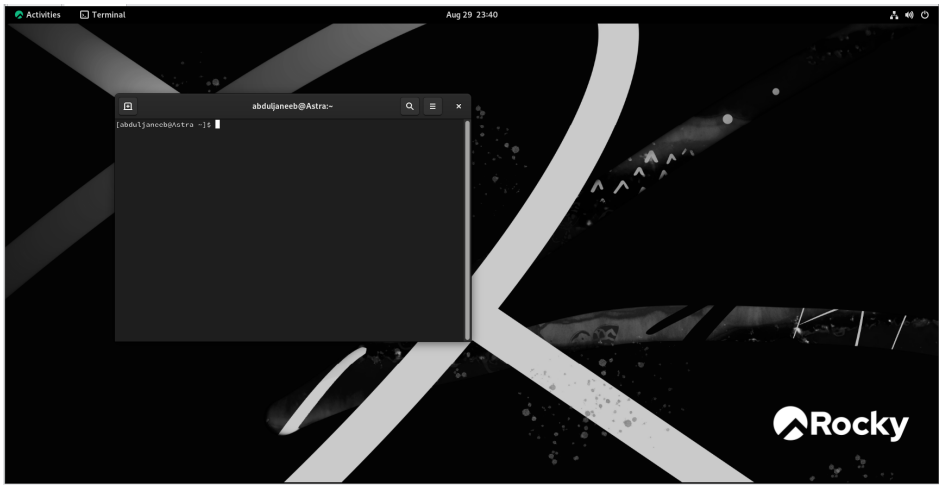
```
    unzip-6.0-60.el9_8.x86_64
    urw-base35-c059-fonts-20200910-6.el9.0.1.noarch
    urw-base35-fonts-20200910-6.el9.0.1.noarch
    urw-base35-gothic-fonts-20200910-6.el9.0.1.noarch
    urw-base35-nimbus-roman-fonts-20200910-6.el9.0.1.noarch
    urw-base35-p052-fonts-20200910-6.el9.0.1.noarch
    urw-base35-z003-fonts-20200910-6.el9.0.1.noarch
    usb_modeswitch-data-20191128-6.el9.0.1.noarch
    util-linux-user-2.37.4-25.el9.x86_64
    virt-what-1.27-2.el9.x86_64
    vte-profile-0.64.2-2.el9.x86_64
    vulkan-loader-1.4.328.1-1.el9.x86_64
    webkit2gtk3-2.52.5-1.el9_8.x86_64
    wireless-regdb-2025.10.07-1.el9.noarch
    wpa_supplicant-1:2.11-2.el9.x86_64
    xdg-dbus-proxy-0.1.3-1.el9.x86_64
    xdg-user-dirs-0.17-10.el9.x86_64
    xdg-utils-1.1.3-13.el9_6.noarch
    xkbcomp-1.4.4-4.el9.x86_64
    xorg-x11-drv-fbdev-0.5.0-11.el9.x86_64
    xorg-x11-drv-vmware-13.2.1-19.el9.x86_64
    xorg-x11-drv-wacom-serial-support-1.0.0-5.el9.x86_64
    xorg-x11-server-Xwayland-24.1.9-4.el9_8.3.x86_64
    xorg-x11-server-utils-7.7-44.el9.x86_64
    xorg-x11-xauth-1:1.1-10.el9.x86_64
    xorg-x11-xinit-session-1.4.0-11.el9.x86_64
    yelp-2:40.3-3.el9_8.1.x86_64
    yelp-tools-40.0-3.el9.0.1.noarch
    zenity-3.32.0-8.el9.x86_64
    urw-base35-bookman-fonts-20200910-6.el9.0.1.noarch
    urw-base35-d050001-fonts-20200910-6.el9.0.1.noarch
    urw-base35-fonts-common-20200910-6.el9.0.1.noarch
    urw-base35-nimbus-mono-ps-fonts-20200910-6.el9.0.1.noarch
    urw-base35-nimbus-sans-fonts-20200910-6.el9.0.1.noarch
    urw-base35-standard-symbols-ps-fonts-20200910-6.el9.0.1.noarch
    usb_modeswitch-2.6.1-4.el9.x86_64
    usbutils-018-1.el9.x86_64
    vdo-8.2.2.2-1.el9.x86_64
    volume_key-libs-0.3.12-17.el9.x86_64
    vte291-0.64.2-2.el9.x86_64
    wavpack-5.4.0-5.el9.x86_64
    wget-1.21.1-8.el9_4.x86_64
    words-3.0-39.el9.0.1.noarch
    xcb-util-0.4.0-19.el9.x86_64
    xdg-desktop-portal-gnome-41.2-3.el9.x86_64
    xdg-user-dirs-gtk-0.10-22.el9.x86_64
    xfsdump-3.1.12-4.el9_3.x86_64
    xorg-x11-drv-evdev-2.10.6-12.el9.x86_64
    xorg-x11-drv-libinput-1.0.1-4.el9_6.x86_64
    xorg-x11-drv-wacom-1.0.0-5.el9.x86_64
    xorg-x11-server-Xorg-1.20.11-34.el9_8.3.x86_64
    xorg-x11-server-common-1.20.11-34.el9_8.3.x86_64
    xorg-x11-utils-7.5-40.el9.x86_64
    xorg-x11-xinit-1.4.0-11.el9.x86_64
    yajl-2.1.0-25.el9.x86_64
    yelp-libs-2:40.3-3.el9_8.1.x86_64
    yelp-xsl-40.2-1.el9_6.1.noarch
    zip-3.0-35.el9.x86_64

  Complete!
```

I set the default target to graphical.target and rebooted. The login screen came up fine with abduljaneeb listed, and I could log in through the GUI – both the desktop and a terminal worked.

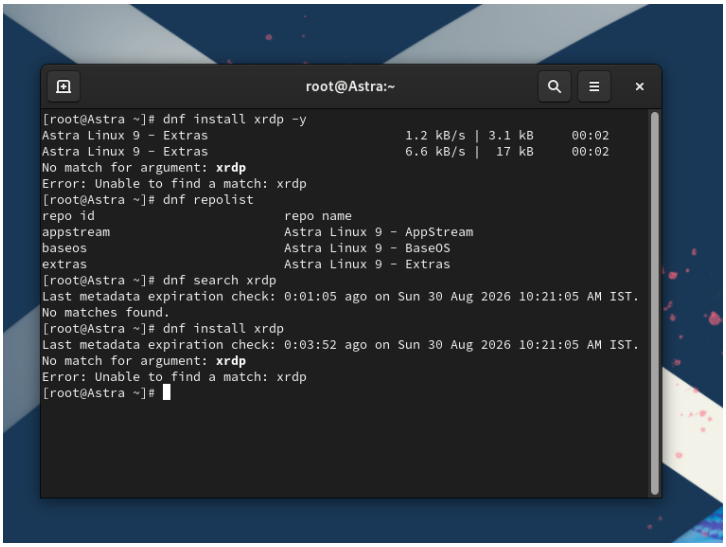


Here's the desktop after logging in through the GUI, with a terminal open.

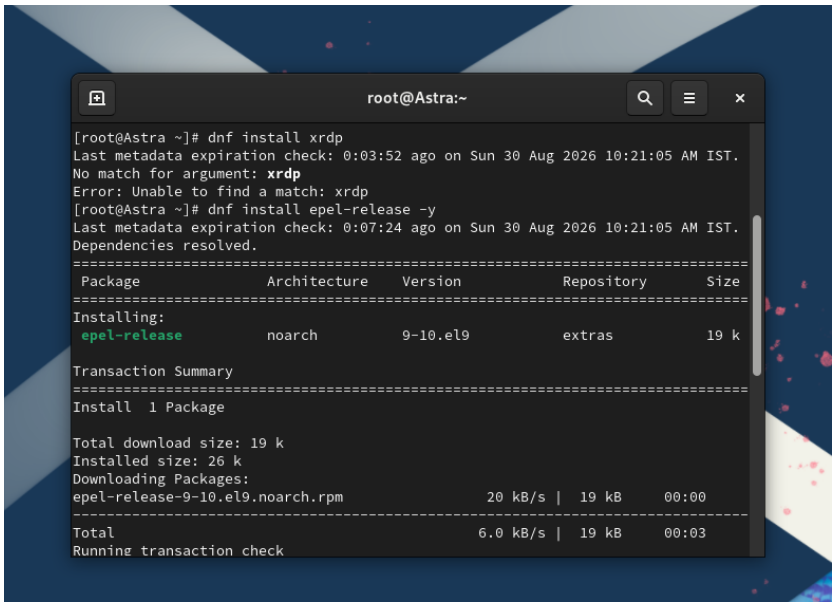


Task 5: Remote Desktop Access

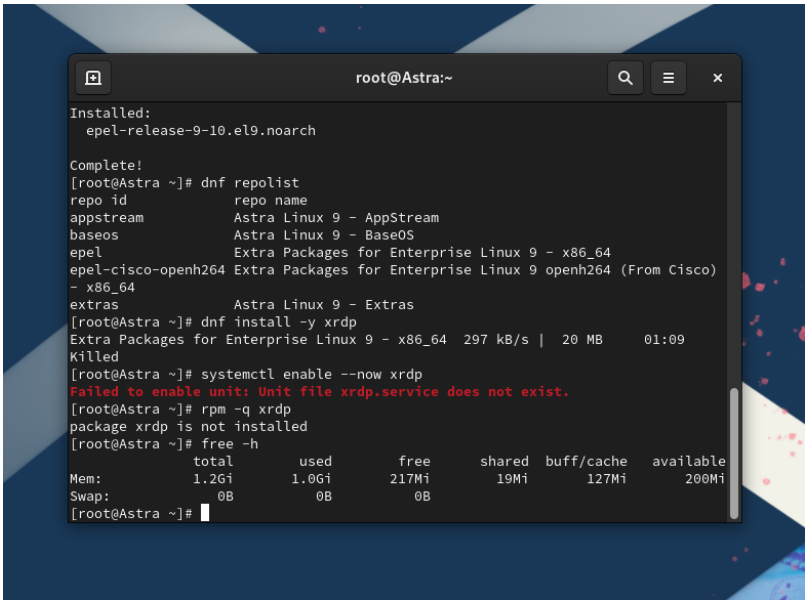
I tried dnf install xrdp -y but it failed with no match found. I checked dnf repolist and only AppStream, BaseOS and Extras were enabled – no EPEL. dnf search xrdp also came back empty, so I installed the EPEL repo with dnf install epel-release -y.



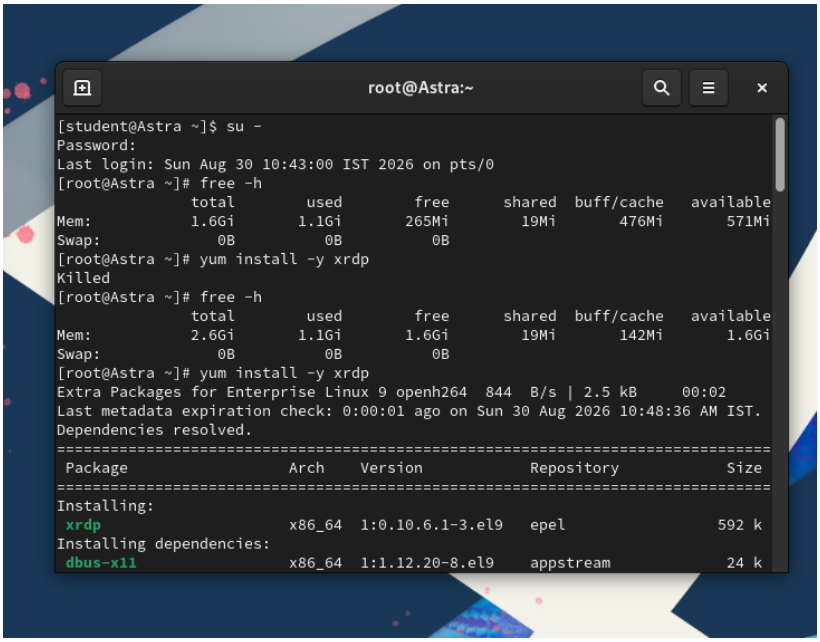
EPEL installed fine and the repolist now showed epel and epel-cisco-openh264.



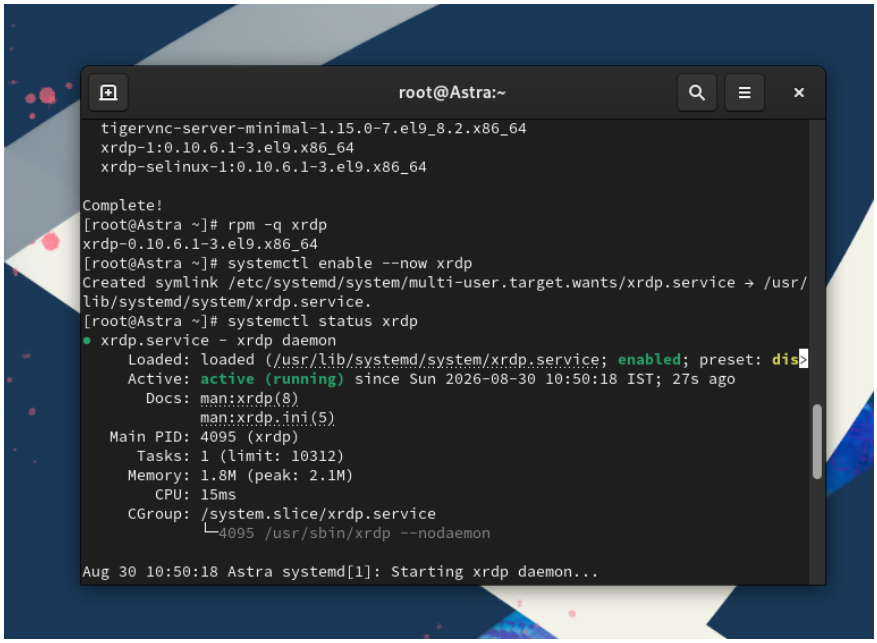
I tried dnf install -y xrdp again, but the process got Killed halfway through. I checked with free -h and saw only 200Mi available, so it looked like a memory issue.



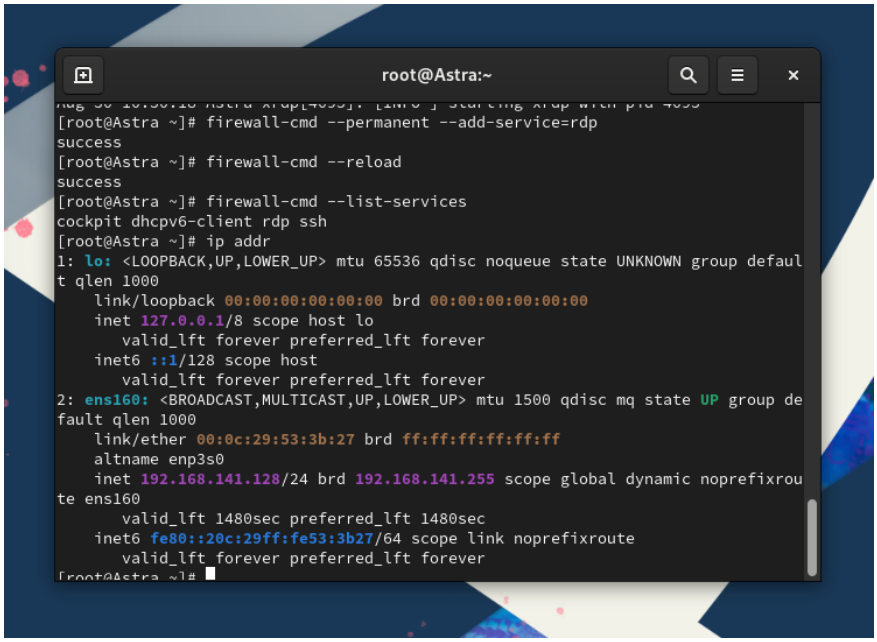
I increased the VM's memory and tried again – free -h now showed a lot more RAM available. I ran yum install -y xrdp again and it started installing xrdp and its dependencies.



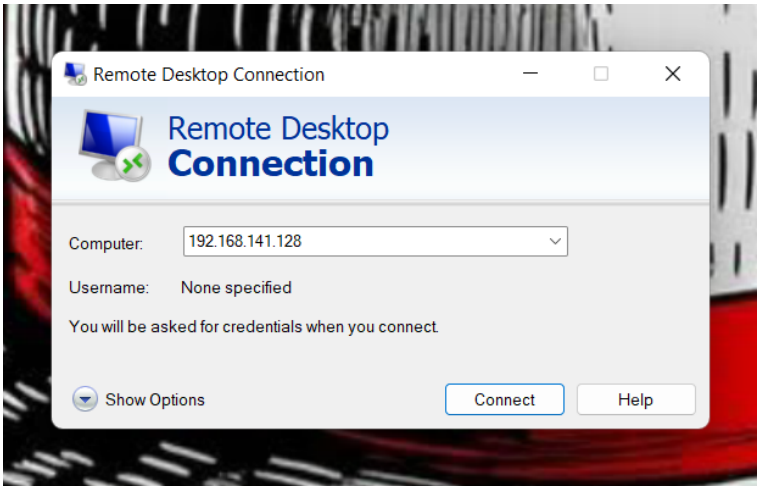
This time it finished. I checked it with rpm -q xrdp, enabled and started the service with systemctl enable --now xrdp, and checked the status – active and running.



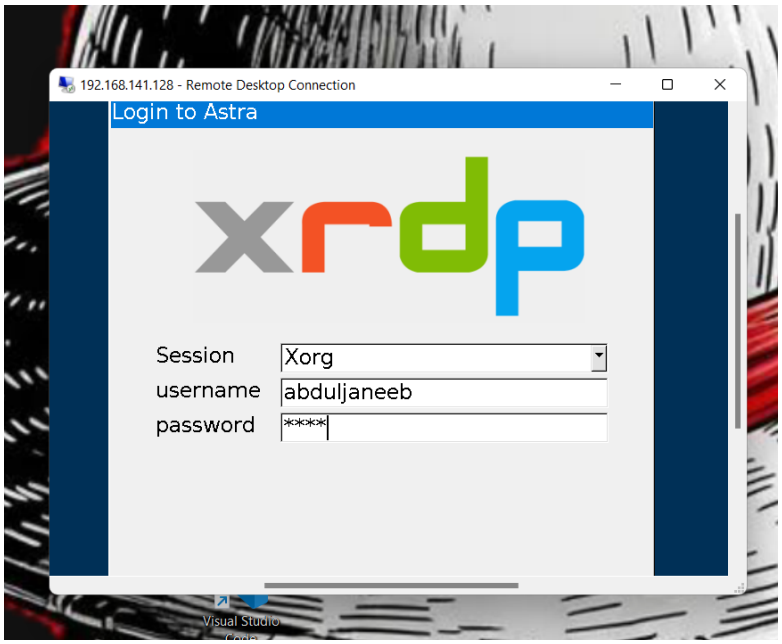
I opened the firewall for RDP with firewall-cmd --permanent --add-service=rdp, reloaded it, and noted down the VM's IP address.



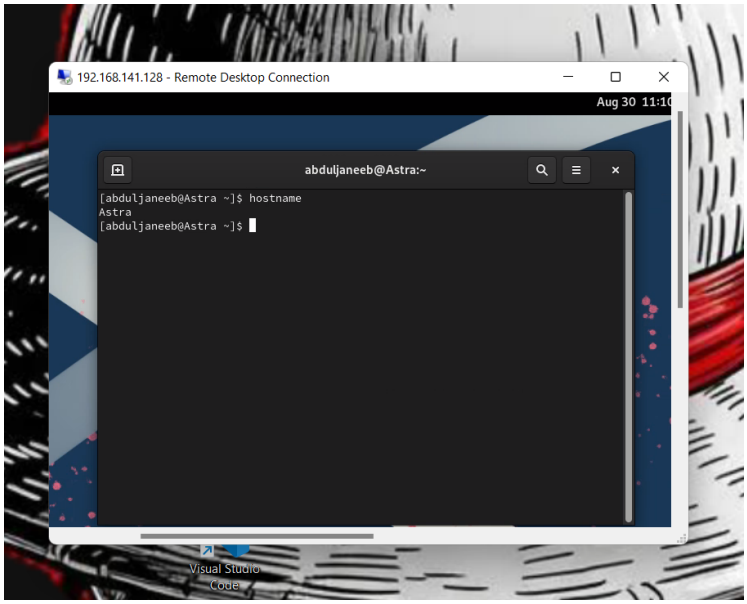
From the Windows host I opened Remote Desktop Connection and entered the VM's IP.



The xrdp login screen came up, so I entered the username abduljaneeb and the password.



It connected – I could see the full Astra desktop remotely from the Windows host, with the terminal working too.



Thank you